

Andrew Phengthalasy

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Education

Worcester Polytechnic Institute

Worcester, MA

Bachelor of Science: Computer Science | **Minor(s):** Financial Technology & Data Science

May 2028

Relevant Coursework: Data Modeling & Analysis, Financial Analytics, Software Engineering, Computational Methods, Database Systems I, Algorithms, Systems Programming Concepts, Object Oriented Design Concepts

Skills

Programming & Web: Python, Typescript, SQL, HTML/CSS, Java, R, C/C++, NodeJS, Javascript, React, Next.js, FastAPI

DevOps: Git, Jira, Agile/Scrum, Auth0, Supabase, PrismaORM, Docker, Nginx, Postman, Power Automate, Render, Vercel

Data & AI: LangChain, Matplotlib, Seaborn, Scikit-learn, Statsmodels, Mistral AI, IBM Watsonx & Granite, IBM Bob

Projects

NarrateAI | IBM AI Builders Challenge - Global Hackathon | *IBM Watsonx & Granite, IBM Bob*

- Architected a full-stack AI web application delivering **12** narrative combinations across **4 audience modes & tones** E2E
- Designed a **2 step LangChain Pipeline** focusing on data ingestion, **narrative synthesis**, and prompt chaining
- Leveraged IBM Granite via **Watsonx API** with retry logic & rate limit handling using **3 modular** Python parsers

Content Management System | *Hanover Insurance Collaboration* | *Agile/Scrum, PERN*

- Refactored backend architecture & integrated **50+ API calls** into multiple employee management pages, tested via **Postman**
- Integrated a **Mistral AI-powered Chatbot**, parsing **300+ documents & 10+ file** types across all Supabase buckets
- Implemented accessibility features utilizing **I18next**, translating UI into **10 unique** languages
- Collaborated on a **10 person Agile** team to deliver a full-stack document, user, workflow, and role management system

Sector Returns During Recessions Analysis | *Python, Pandas, FRED API, Yahoo Finance*

- Built an **E2E ETL Pipeline** to scrape, clean & align **324 months** of sector-level excess returns, **NBER** recession dummies, & macroeconomic **INDPRO, MKT** controls
- Executed Newey-West **HAC OLS Models (12 lags)**, via statsmodels producing market betas from **0.27** to **1.27** with strong statistical significance
- Delivered model diagnostics confirming **recession coefficient $p > 0.05$** across six sectors, rejecting the hypothesis

Housing Price Prediction Model | *Python, Scikit-learn, Pandas, Seaborn*

- Built a **preprocessing pipeline** for **545+ records**, converting categorical variables into quantitative data
- Implemented **Simple Linear, Multiple Linear, and Lasso Regression**, resulting in an **89%** improvement in accuracy
- Quantified model performance via **R²** and **MSE** to conclude feasibility of price predictions

Experience

Research Assistant | *Research Solutions Institute (WPI)*

April 2025 - Present

- Maintain a centralized **Spin Based Opportunity** database (**infoEd**) to streamline management of **100+ faculty research** proposals and grant proposals.
- Automated administrative workflows by designing standardized templates, reducing proposal preparation time by **15%**
- Collaborate with RSI team to identify and implement **software solutions** to improve research efficiency by **25%**

Fundraising Intern | *Society of Asian Scientists & Engineers (WPI)*

March 2026 - May 2026

- Strategized **3+** meaningful fundraising efforts, raising **3 times** the annual amount of the previous year after only **3 months**
- Partnered with **6 Executive Board** members to streamline event logistics
- Analyzed **fundraising data** to gain **valuable insights** for continuous improvement in future efforts

Involvements

Layout & Logistics Manager | *Pan Asian Association*

September 2025 - May 2026

PASS-CS Scholar | *NSF & Computer Science Department*

August 2024 - May 2028

Student Worker | *Housing & Residential Experience Center*

September 2024 - Present